

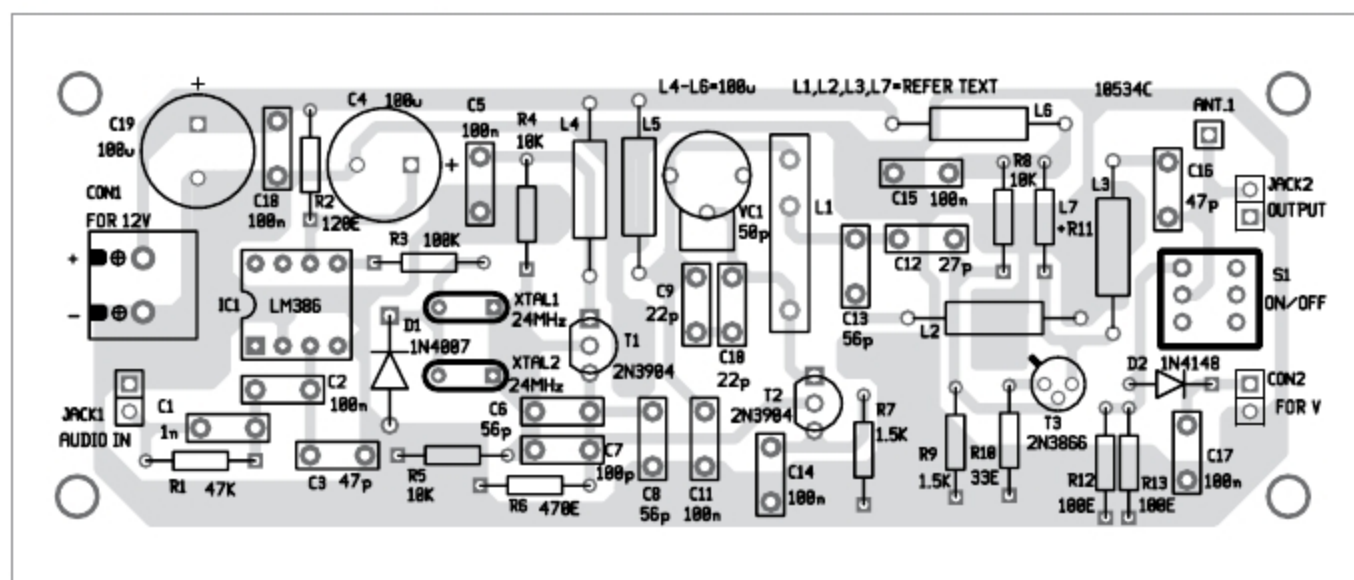


Fig. 3: Components layout for the PCB

make a series-tuned (72MHz) harmonics trap and attenuate the strong third harmonic to ground. Ready-made 100µH moulded inductors (L4 through L6) isolate the power supply from transistors T1 through T3 to

help minimise interference between different stages.

T2 feeds power amplifier T3 with approximately 5mW of RF drive via 1/4-turn tap from the top of tuning coil L1. T3 is biased for class A operation. Standing current is set to 20mA with resistors R8, R9 and R10.

40mW of RF power is available at collector of T3. Matching network comprising L3 and C16 transforms output impedance of T3 to 50-ohm impedance of the antenna. (A 78cm long single-strand wire was used as an antenna during testing.)

Audio amplifier IC1's frequency modulates oscillator T1 by varying capacitance of diode D1, which is connected in series with XTAL1 and XTAL2. (Note that XTAL1 and XTAL2 are connected in parallel to have greater frequency deviation.) The transmitter accepts audio input from an external source such as cellphone. External volume level should be set for best audio level output. R1 and C1 add 50µs pre-emphasis to boost higher audio frequencies. Increasing R1 to 15-kilo-ohm changes pre-emphasis to 75µs.

Construction and testing

An actual-size PCB layout of the FM transmitter is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect 12V DC across CON1.

Keep all leads short. The prototype was built on a piece of copper-clad board using ugly bug construction method. The large ground plane of the copper board gives excellent results for the VHF circuit. With S1 on and 50-ohm dummy load (R12

and R13) connected, apply 12V regulated power supply.

Carefully tune VC1 for maximum voltage (V) reading at CON2. Output power in milliwatts equals

$(V + 0.5V \text{ diode loss})^2 / 100$

No further adjustment is required.

A cheap USB SDR dongle and free spectrum analyser software may be used to observe output. It can be downloaded from <https://www.rtl-sdr.com/tag/spectrum-analyzer-2/>

Fine tuning of VC1 can also be done using a handheld FM receiver. Make sure a wire antenna is connected in the transmitter circuit.

Now, with FM receiver in your hand, slowly walk away from the transmitter. Correct tuning of VC1 will give maximum range on 96MHz. T3 can be replaced with 2N3904. In this case, increase R10 to 56-ohm. Power output will drop to 15mW.

Use a suitable heat-sink for 2N3866. Care should be taken to keep L1 and L2 2cm apart and at ninety degrees to each other on the PCB.

Spectrum output of the transmitter obtained is in the range of 24MHz to 700MHz. Second and third harmonics of 96MHz are down to -18dBc and -24dBc, respectively. All other spurs or distortions are below -20dBc.

Switch S1 should be off when antenna is connected. Use a shielded cable for input audio connection.

Some inductors are self-made, while others are readymade. L1 is a three-turn 20SWG copper-enameled coil on 8mm diameter air core, tapped 1/4 turns from positive side. L2 also is a three-turn 20SWG copper-enameled wire on 8mm diameter air core. L3 is seven-turn 20SWG copper-enameled wire on 8mm diameter air core. L4 through L6 are 100µH fixed-value coils.

Inductor L7 is made of 70 turns of 41SWG copper enameled wire on resistor R11. **EFY**



Joy Mukherji is an electronics hobbyist who likes designing radio frequency circuits

PARTS LIST

Semiconductors:

IC1	- LM386 audio amplifier
T1, T2	- 2N3904 transistor
T3	- 2N3866 transistor
D1	- 1N4007 rectifier diode
D2	- 1N4148 signal diode

Resistors (all 1/4-watt, ±5% carbon):

R1	- 47-kilo-ohm
R2	- 120-ohm
R3	- 100-kilo-ohm
R4, R5, R8, R11	- 10-kilo-ohm
R6	- 470-ohm
R7, R9	- 1.5-kilo-ohm
R10	- 33-ohm
R12, R13	- 100-ohm

Capacitors:

C1	- 1nF ceramic disk
C2, C5, C11, C14, C15, C17, C18	- 100nF ceramic disk
C3	- 47pF ceramic disk
C4, C19	- 100µF, 35V electrolytic
C6, C8, C13	- 56pF ceramic disk
C7	- 100pF ceramic disk
C9, C10	- 22pF ceramic disk
C12	- 27pF ceramic disk
C16	- 47pF ceramic disk

Miscellaneous:

CON1	- 2-pin terminal connector
CON2	- 2-pin connector
Jack1, Jack2	- 2-pin connector with mono jack
XTAL1, XTAL2	- 24MHz crystal
S1	- On/off switch
L1, L2, L3, L7	- Refer text/Fig. 1
L4-L6	- 100µH fixed-value coil
ANT.1	- 78cm single strand antenna wire

- 8-pin IC base
- 12V DC regulated power supply
- Heat-sink for 2N3866
(All these are available from kitsnspares.com)